

Manwe Castro

manwe123@asu.edu | (602) 350-3164 | linkedin.com/in/manwe-castro | https://manwe-123.github.io/

Education

Bachelor of Science in Mechanical Engineering

Arizona State University | GPA: 3.50

Tempe, AZ

Expected May 2028

Relevant Coursework: Mechanics of Materials, Structure and Properties of Materials, Statics, Thermodynamics, Dynamics, Circuits

Project Experience

Mental Health Robotics System for Isolated Environments

Tempe, AZ

Design and Prototyping Lead

August 2024 – December 2024

- Programmed an Arduino Uno microcontroller to process ultrasonic and photoresistive sensor inputs, controlling drive motors and a servo actuator to enable autonomous user tracking and automated medical dispensing, fulfilling the core requirement for interactive companion functionality.
- Modeled the robotic enclosure in CAD and manufactured the structure using laser-cut balsa and plywood, optimizing for lightweight durability and precise mechanical fit to successfully house all sensory and actuation components under strict material constraints.
- Executed formal acceptance testing to verify sensor accuracy and motor response, diagnosing and resolving electrical power delivery issues during integration to successfully validate the semi-autonomous navigation and automated dispensing mechanisms.

VR Rapid Prototyping Application

Tempe, AZ

Prototyping Lead | Modeling Team

March 2025 – December 2025

- Engineered a virtual reality rapid prototyping environment for Meta Quest headsets using Unity and C#, integrating over 100 interactive 3D assets with simulated physics and collision mechanics to provide an immersive platform for undergraduate engineering design exercises.
- Directed a cross-functional modeling team by establishing standardized Unity asset workflows and resolving engine-level rendering blockers, optimizing frame rates to mitigate VR motion sickness and ensuring the on-time delivery of a stable, functional prototype.
- Collaborated with faculty to integrate the VR prototype into the FSE100 curriculum and developed in-game analytics scripts to track user interactions and completion times, creating a data-driven feedback loop that enables instructors to quantify student concept mastery.

Documentation Lead

January 2025 – March 2025

- Authored comprehensive technical documentation, including Gantt charts and risk analysis matrices, to track design milestones and maintain version control across a multi-semester project lifecycle.
- Standardized reporting templates and delivered a final technical transition package detailing system architecture, hardware deployment logistics, and future development recommendations for subsequent engineering teams.

Shock Absorber Assembly

August 2025 – December 2025

- Designed a fully constrained automotive shock absorber assembly in SolidWorks by modeling custom components, integrating McMaster-Carr standard hardware, and amending initial dimensions to eliminate physical interferences, ensuring proper mechanical fit and material selection for dynamic suspension applications.
- Generated fully dimensioned 2D engineering drawings following ASME Y14.5 standards and 3rd angle projection, detailing complex features such as shaft thread specifications, keyways, and spring parameters to provide machinists with precise, unambiguous fabrication instructions.
- Authored a comprehensive Bill of Materials and exploded assembly drawing to document custom and standard hardware components, establishing a standardized technical reference that streamlines procurement and future mechanical assembly.

Leadership and Campus Involvement

Society of Hispanic Professional Engineers

Tempe, AZ

Outreach Director of STEM Events

March 2026 – Present

- Direct chapter outreach and design technical workshops to build practical engineering skills for undergraduate members.
- Develop the fall semester event calendar and workshop curriculum, focusing on hands-on technical training and professional development.

Professional Experience

Institute for Digital Inclusion Acceleration

Chandler, AZ

Digital Navigator

January 2025 – Present

- Instruct over 300 weekly visitors in hardware and software fundamentals, facilitating thousands of hands-on technology interactions across diverse age groups.
- Develop and deliver structured curriculum for nine distinct software platforms, including Swift, TinkerCAD, and SpheroEDU, standardizing the instructional process for the center.
- Operate and maintain Flashforge 3D printers using Orca Slicer, troubleshooting hardware failures and optimizing print parameters to ensure consistent output quality.

Skills

CAD & Design: SolidWorks, Fusion 360, 2D Drafting, MATLAB

Fabrication & Hands-On: FDM 3D Printing, Circuit Wiring, Laser Cutting, Orca Slicer, Automotive Repair, Construction,

Programming & Controls: Python, Arduino IDE (C++), Unity Game Engine, Git, Sensor Integration